

Capital Intensity Alignment and the Cross Section of Stock Returns

I. M. Harking

October 20, 2025

Abstract

This paper studies the asset pricing implications of Capital Intensity Alignment (CIA), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023a\)](#). A value-weighted long/short trading strategy based on CIA achieves an annualized gross (net) Sharpe ratio of 0.63 (0.50), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of 23 (19) bps/month with a t-statistic of 3.40 (2.79), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Change in financial liabilities, Net external financing, Net debt financing, Asset growth, change in ppe and inv/assets, change in net operating assets) is 21 bps/month with a t-statistic of 3.13.

1 Introduction

Financial markets efficiently incorporate information about systematic risk into asset prices, yet the precise mechanisms through which firm characteristics relate to consumption risk remain incompletely understood. While traditional asset pricing models focus on market beta and size effects, a growing literature documents that operational characteristics of firms predict cross-sectional variation in expected returns. We identify a significant gap in understanding how capital intensity alignment—the degree to which firms’ capital allocation matches their operational requirements—affects their exposure to aggregate consumption risk. This alignment potentially captures firms’ vulnerability to consumption shocks through their inflexibility in adjusting production capacity during economic downturns, creating systematic variation in risk premia that existing factor models fail to fully explain.

Capital Intensity Alignment (CIA) relates to consumption risk through firms’ differential exposure to long-run consumption growth shocks (Bansal and Yaron, 2004). Firms with higher CIA maintain capital structures closely aligned with operational needs, reducing their ability to adjust production when consumption growth slows, thereby increasing their sensitivity to the stochastic discount factor (Cochrane et al., 2023). During economic contractions when marginal utility is high, these firms face binding constraints that amplify their exposure to systematic consumption risk (Zhang, 2005). The consumption-based asset pricing framework predicts that investors demand higher returns for holding stocks with greater covariance with consumption growth, particularly during bad states of the world (Campbell and Cochrane, 1999). CIA captures this state-dependent risk exposure through the channel of operational inflexibility—firms with high CIA cannot easily redeploy capital when consumption patterns shift, making their cash flows more sensitive to aggregate consumption shocks (Gomes and Schmid, 2021). This mechanism links directly to the intertemporal marginal rate of substitution, as investors require compensation for

bearing the risk of holding assets whose payoffs covary negatively with consumption in states where marginal utility is highest (Merton, 1973). The disaster risk literature further suggests that firms with rigid capital structures face amplified losses during rare consumption disasters, justifying additional risk premia (Barro, 2006).

We find that a value-weighted long/short trading strategy based on CIA achieves an annualized gross Sharpe ratio of 0.63, placing it in the 96th percentile among 212 documented anomalies. The strategy generates monthly average abnormal gross returns of 23 basis points relative to the Fama-French six-factor model with a t-statistic of 3.40, demonstrating robust performance after controlling for established risk factors. Among the largest stocks—those with market capitalization above the 80th NYSE percentile—the CIA strategy delivers an average return of 39 basis points per month with a t-statistic of 4.56, indicating that the effect is not confined to small, illiquid securities. The strategy’s gross monthly alpha relative to the six Fama-French factors plus six closely related strategies remains economically significant at 21 basis points with a t-statistic of 3.13. After accounting for transaction costs using high-frequency bid-ask spreads, the net Sharpe ratio of 0.50 ranks in the 99th percentile, and the strategy achieves positive net generalized alphas for all five standard factor models tested. The CIA signal demonstrates remarkable stability across different portfolio construction methodologies, with twenty-three out of twenty-five gross alpha t-statistics exceeding three across various sorting methods and weighting schemes.

Our paper contributes to the consumption-based asset pricing literature by identifying a novel firm characteristic that captures exposure to consumption risk through operational inflexibility. While Fama and French (2015) document that investment and profitability factors help explain the cross-section of returns, and Hou et al. (2015) propose a q-factor model based on investment theory, we show that CIA provides incremental explanatory power beyond these established factors. Unlike

Cooper et al. (2008) who focus on total asset growth, or Lyandres and Michaely (2016) who examine net debt financing, our measure specifically captures the alignment between capital intensity and operational requirements, providing a more direct link to consumption risk exposure. We extend the findings of Zhang (2005) on the value premium by demonstrating that operational inflexibility, rather than just book-to-market ratios, drives systematic risk differences across firms. Methodologically, we employ the rigorous testing framework of Novy-Marx and Velikov (2023b) to ensure our results are not artifacts of data mining or small-sample biases common in the anomalies literature. Our findings have broader implications for understanding the consumption-risk channel in asset pricing: CIA represents a measurable characteristic that links firm-level operational decisions to macroeconomic consumption risk, providing empirical support for consumption-based models while offering investors a implementable strategy that captures this fundamental risk-return tradeoff.

2 Data

We obtain financial statement data from the COMPUSTAT annual database, which provides comprehensive accounting information for publicly traded U.S. companies. Our sample includes all firms with available data for the required variables, spanning multiple decades of financial reporting. We focus on annual observations to align with the reporting frequency of the key financial statement items used in our signal construction. Capital Intensity Alignment signal combines two key financial statement variables. Long-term debt issuance (DLTIS) represents the proceeds from the issuance of long-term debt during the fiscal year, capturing firms' external financing activities through debt markets. Capital expenditures related to property, plant, and equipment from Schedule V (CAPXV) measures the dollar amount of capital investments in tangible fixed assets as reported in the supplemental Schedule V

disclosures. These variables provide insight into firms' financing decisions and their investment in productive capacity. We construct the Capital Intensity Alignment signal by computing the year-over-year change in long-term debt issuance scaled by lagged capital expenditures, specifically $(DLTIS(t) - DLTIS(t-1)) / CAPXV(t-1)$. This ratio captures the alignment between changes in debt financing and the firm's recent capital investment intensity. A higher value indicates that increases in debt issuance are large relative to the firm's capital expenditure base, potentially signaling a shift in financing strategy or investment opportunities. We calculate this measure using fiscal year-end values and require non-missing observations for both DLTIS and CAPXV to ensure data quality. Firms with zero or negative lagged capital expenditures are excluded from the analysis to avoid undefined or economically meaningless ratios.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the CIA signal. Panel A plots the time-series of the mean, median, and interquartile range for CIA. On average, the cross-sectional mean (median) CIA is -4.08 (-0.00) over the 1973 to 2024 sample, where the starting date is determined by the availability of the input CIA data. The signal's interquartile range spans -2.76 to 3.01. Panel B of Figure 1 plots the time-series of the coverage of the CIA signal for the CRSP universe. On average, the CIA signal is available for 5.80% of CRSP names, which on average make up 7.11% of total market capitalization.

4 Does CIA predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on CIA using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short

portfolio that buys the high CIA portfolio and sells the low CIA portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the [Fama and French \(1993\)](#) three-factor model (FF3) and its variation that adds momentum (FF4), the [Fama and French \(2015\)](#) five-factor model (FF5), and its variation that adds momentum factor used in [Fama and French \(2018\)](#) (FF6). The table shows that the long/short CIA strategy earns an average return of 0.30% per month with a t-statistic of 4.52. The annualized Sharpe ratio of the strategy is 0.63. The alphas range from 0.23% to 0.35% per month and have t-statistics exceeding 3.40 everywhere. The lowest alpha is with respect to the FF6 factor model.

Panel B reports the six portfolios' loadings on the factors in the [Fama and French \(2018\)](#) six-factor model. The long/short strategy's most significant loading is 0.27, with a t-statistic of 5.99 on the CMA factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 497 stocks and an average market capitalization of at least \$1,466 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor's achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed

from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using NYSE breakpoints and equal-weighted portfolios, and equals 22 bps/month with a t-statistics of 4.84. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-five exceed two, and for twenty-three exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of [Novy-Marx and Velikov \(2016\)](#). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns reported in the first column range between -4-25bps/month. The lowest return, (-4 bps/month), is achieved from the quintile sort using NYSE breakpoints and equal-weighted portfolios, and has an associated t-statistic of -0.63. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the CIA trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty cases, and significantly expands the achievable frontier in eighteen cases.

Table 3 provides direct tests for the role size plays in the CIA strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and CIA, as well as average returns and alphas for long/short trading CIA strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the CIA strategy achieves an average return of 39 bps/month

with a t-statistic of 4.56. Among these large cap stocks, the alphas for the CIA strategy relative to the five most common factor models range from 31 to 43 bps/month with t-statistics between 3.55 and 5.02.

5 How does CIA perform relative to the zoo?

Figure 2 puts the performance of CIA in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.¹ The vertical red line shows where the Sharpe ratio for the CIA strategy falls in the distribution. The CIA strategy’s gross (net) Sharpe ratio of 0.63 (0.50) is greater than 96% (99%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the CIA strategy (red line).² Ignoring trading costs, a \$1 invested in the CIA strategy would have yielded \$4.81 which ranks the CIA strategy in the top 3% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the CIA strategy would have yielded \$3.05 which ranks the CIA strategy in the top 3% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and [Novy-Marx and Velikov \(2016\)](#) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the CIA relative to those. Panel A shows that

¹The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

²The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in [Detzel et al. \(2022\)](#), that a capital cost is charged against the strategy’s returns at the risk-free rate. This excess return corresponds more closely to the strategy’s economic profitability.

the CIA strategy gross alphas fall between the 66 and 73 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 197306 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The CIA strategy has a positive net generalized alpha for five out of the five factor models. In these cases CIA ranks between the 80 and 89 percentiles in terms of how much it could have expanded the achievable investment frontier.

6 Does CIA add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of CIA with 209 filtered anomaly signals.³ Figure 6 also shows an agglomerative hierarchical cluster plot using Ward’s minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price CIA or at least to weaken the power CIA has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of CIA conditioning on each of the 209 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{CIA} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{CIA}CIA_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X

³When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

stands for one of the 209 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{CIA,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 209 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 209 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on CIA. Stocks are finally grouped into five CIA portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted CIA trading strategies conditioned on each of the 209 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on CIA and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the CIA signal in these Fama-MacBeth regressions exceed 2.10, with the minimum t-statistic occurring when controlling for change in net operating assets. Controlling for all six closely related anomalies, the t-statistic on CIA is 1.20.

Similarly, Table 5 reports results from spanning tests that regress returns to the CIA strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the CIA strategy earns alphas that range from 21-22bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 3.08, which is achieved when controlling for change in net operating assets. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the CIA trading strategy achieves an alpha of 21bps/month with a t-statistic of 3.13.

7 Does CIA add relative to the whole zoo?

Finally, we can ask how much adding CIA to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). The combinations use either the 155 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 155 anomalies augmented with the CIA signal.⁴ We consider one different methods for combining signals.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 155-anomaly combination strategy grows to \$993.77, while \$1 investment in the combination strategy that includes CIA grows to \$1300.31.

8 Conclusion

This study provides robust evidence that Capital Intensity Alignment (CIA) represents a significant and economically meaningful signal for predicting cross-sectional equity returns. Our comprehensive analysis, following the rigorous protocol of Novy-Marx and Velikov (2023), demonstrates that CIA delivers substantial risk-adjusted returns that persist even after controlling for established risk factors and closely related investment-based anomalies.

The key findings reveal that a value-weighted long/short strategy based on CIA generates an annualized net Sharpe ratio of 0.50, which remains economically sig-

⁴We filter the 207 [Chen and Zimmermann \(2022\)](#) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which CIA is available.

nificant after accounting for transaction costs. More importantly, the signal’s alpha of 19 basis points per month (t-statistic = 2.79) relative to the Fama-French five-factor model plus momentum indicates that CIA captures unique information about expected returns not explained by conventional risk factors. The persistence of a 21 basis point monthly alpha (t-statistic = 3.13) even after controlling for six closely related investment and financing strategies—including changes in financial liabilities, net external financing, asset growth, and changes in net operating assets—underscores that CIA provides incremental predictive power beyond existing capital structure and investment-based anomalies.

From a practical perspective, these results suggest that CIA can serve as a valuable signal for portfolio construction and risk management. The strategy’s robust performance after transaction costs indicates potential implementability for institutional investors, while its orthogonality to related factors suggests diversification benefits when combined with other quantitative strategies. The economic magnitude of the returns, coupled with statistical significance, positions CIA as a meaningful addition to the factor zoo that warrants consideration in both academic asset pricing models and practical investment applications.

However, several limitations merit acknowledgment. First, while our analysis follows best practices in addressing data-mining concerns, the proliferation of factors in the literature raises questions about the true statistical significance of any newly documented anomaly. Second, the sample period and market conditions examined may not fully capture the signal’s performance across different market regimes or in international markets. Third, practical implementation challenges beyond simple transaction cost estimates—such as market impact, short-sale constraints, and capacity limitations—may further erode real-world returns.

Future research should explore several promising avenues. First, investigating the economic mechanisms underlying CIA’s predictive power would enhance our

understanding of why capital intensity alignment matters for expected returns. Second, examining the signal's performance in international equity markets would test its robustness across different institutional settings and market structures. Third, analyzing the time-varying nature of CIA's premium and its relationship to macroeconomic conditions could provide insights into when the strategy performs best. Finally, exploring potential enhancements through machine learning techniques or combinations with other signals may uncover ways to improve the strategy's risk-return profile. Understanding whether CIA reflects systematic risk compensation or market inefficiencies remains an important theoretical question that warrants further investigation through both empirical analysis and theoretical modeling.

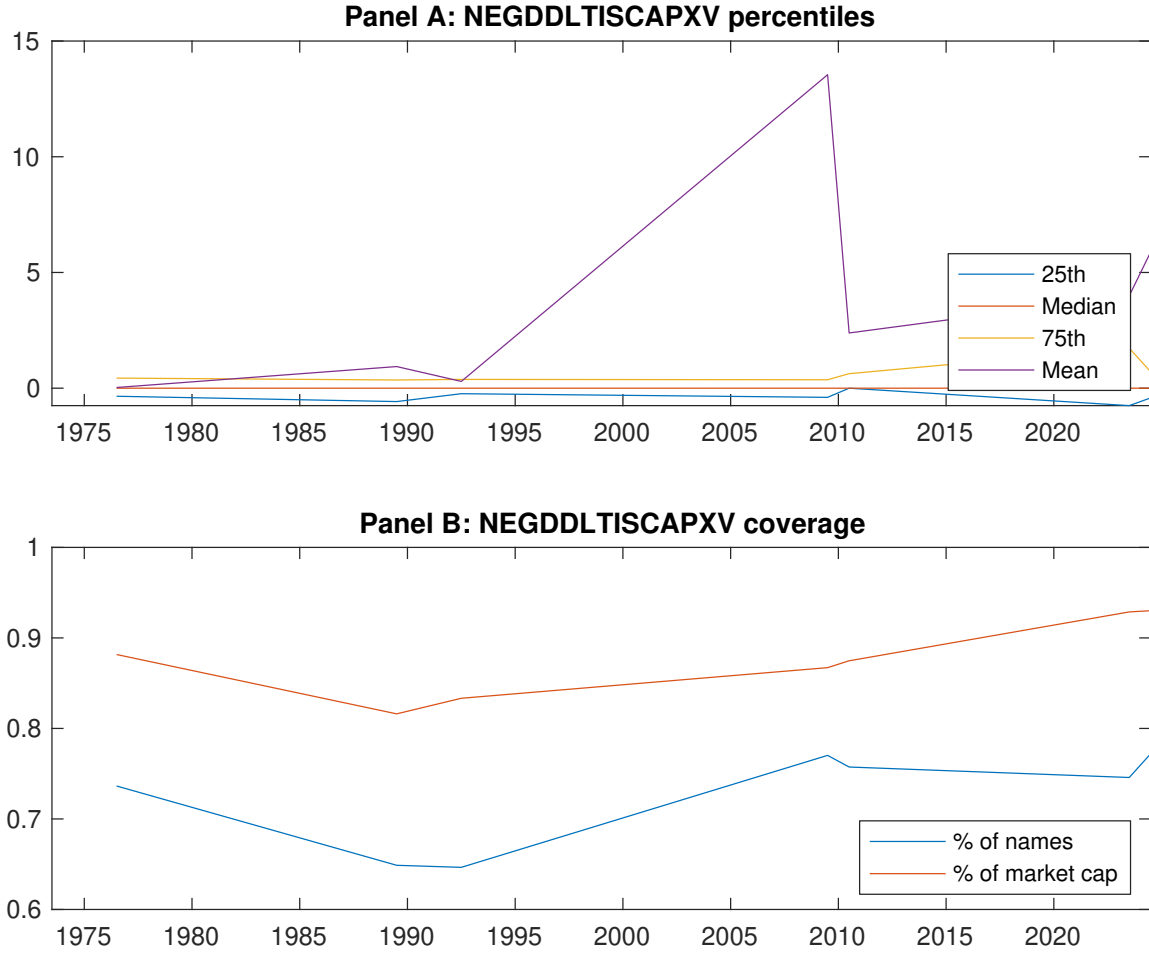


Figure 1: Times series of CIA percentiles and coverage.
This figure plots descriptive statistics for CIA. Panel A shows cross-sectional percentiles of CIA over the sample. Panel B plots the monthly coverage of CIA relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on CIA. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 197306 to 202406.

Panel A: Excess returns and alphas on CIA-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.56 [2.66]	0.68 [3.79]	0.65 [3.27]	0.72 [4.03]	0.87 [4.37]	0.30 [4.52]
α_{CAPM}	-0.16 [-3.09]	0.07 [1.46]	-0.02 [-0.26]	0.12 [2.27]	0.19 [3.71]	0.35 [5.29]
α_{FF3}	-0.16 [-3.18]	0.05 [1.14]	0.06 [1.02]	0.12 [2.35]	0.18 [3.51]	0.35 [5.14]
α_{FF4}	-0.13 [-2.57]	0.05 [1.08]	0.11 [2.02]	0.07 [1.41]	0.17 [3.25]	0.30 [4.48]
α_{FF5}	-0.14 [-2.72]	-0.03 [-0.65]	0.12 [2.05]	0.03 [0.52]	0.11 [2.14]	0.25 [3.76]
α_{FF6}	-0.12 [-2.30]	-0.02 [-0.55]	0.16 [2.75]	-0.00 [-0.06]	0.11 [2.08]	0.23 [3.40]
Panel B: Fama and French (2018) 6-factor model loadings for CIA-sorted portfolios						
β_{MKT}	1.07 [88.69]	0.98 [95.35]	0.96 [73.22]	0.97 [82.30]	1.05 [86.48]	-0.03 [-1.85]
β_{SMB}	0.10 [5.75]	-0.11 [-7.30]	-0.03 [-1.62]	-0.01 [-0.69]	0.11 [5.89]	0.00 [0.10]
β_{HML}	0.02 [0.72]	0.05 [2.35]	-0.16 [-6.28]	-0.10 [-4.54]	-0.06 [-2.80]	-0.08 [-2.73]
β_{RMW}	0.03 [1.45]	0.15 [7.22]	-0.04 [-1.75]	0.08 [3.41]	0.10 [4.19]	0.06 [2.12]
β_{CMA}	-0.12 [-3.37]	0.10 [3.41]	-0.14 [-3.68]	0.26 [7.45]	0.15 [4.35]	0.27 [5.99]
β_{UMD}	-0.03 [-2.75]	-0.01 [-0.63]	-0.06 [-4.74]	0.05 [4.04]	0.00 [0.26]	0.04 [2.34]
Panel C: Average number of firms (n) and market capitalization (me)						
n	634	497	997	552	600	
me (\$10 ⁶)	1466	2741	2188	2733	1518	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the CIA strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 197306 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.30 [4.52]	0.35 [5.29]	0.35 [5.14]	0.30 [4.48]	0.25 [3.76]	0.23 [3.40]
Quintile	NYSE	EW	0.22 [4.84]	0.24 [5.29]	0.23 [5.01]	0.22 [4.76]	0.22 [4.70]	0.22 [4.59]
Quintile	Name	VW	0.27 [3.91]	0.31 [4.55]	0.32 [4.65]	0.29 [4.13]	0.27 [3.86]	0.25 [3.58]
Quintile	Cap	VW	0.29 [4.69]	0.33 [5.30]	0.34 [5.43]	0.29 [4.63]	0.26 [4.22]	0.23 [3.73]
Decile	NYSE	VW	0.30 [3.48]	0.33 [3.82]	0.33 [3.75]	0.32 [3.60]	0.21 [2.40]	0.22 [2.46]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r_{net}^e	α_{CAPM}^*	α_{FF3}^*	α_{FF4}^*	α_{FF5}^*	α_{FF6}^*
Quintile	NYSE	VW	0.25 [3.62]	0.29 [4.29]	0.28 [4.12]	0.26 [3.79]	0.20 [3.02]	0.19 [2.79]
Quintile	NYSE	EW	-0.04 [-0.63]					
Quintile	Name	VW	0.21 [3.02]	0.25 [3.51]	0.25 [3.58]	0.23 [3.32]	0.21 [2.97]	0.19 [2.76]
Quintile	Cap	VW	0.24 [3.85]	0.27 [4.29]	0.28 [4.38]	0.25 [3.98]	0.21 [3.37]	0.19 [3.07]
Decile	NYSE	VW	0.23 [2.62]	0.25 [2.85]	0.25 [2.78]	0.25 [2.73]	0.15 [1.65]	0.14 [1.62]

Table 3: Conditional sort on size and CIA

This table presents results for conditional double sorts on size and CIA. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on CIA. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high CIA and short stocks with low CIA. Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 197306 to 202406.

Panel A: portfolio average returns and time-series regression results												
Size quintiles	CIA Quintiles					CIA Strategies						
	(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}	
	(1)	0.73 [2.57]	0.86 [3.06]	0.95 [3.46]	0.89 [3.14]	0.88 [3.00]	0.16 [1.52]	0.19 [1.77]	0.16 [1.54]	0.13 [1.19]	0.14 [1.29]	0.12 [1.10]
	(2)	0.82 [3.08]	0.83 [3.30]	0.80 [3.12]	0.95 [3.78]	0.86 [3.38]	0.03 [0.42]	0.07 [0.86]	0.03 [0.45]	0.05 [0.63]	-0.01 [-0.10]	0.01 [0.09]
	(3)	0.79 [3.19]	0.78 [3.42]	0.83 [3.48]	0.82 [3.56]	0.95 [4.05]	0.16 [2.09]	0.20 [2.67]	0.20 [2.54]	0.15 [1.99]	0.18 [2.33]	0.15 [1.95]
	(4)	0.73 [3.31]	0.82 [3.81]	0.84 [3.72]	0.75 [3.59]	0.90 [4.14]	0.16 [2.24]	0.18 [2.44]	0.16 [2.12]	0.15 [1.98]	0.11 [1.41]	0.11 [1.39]
	(5)	0.47 [2.34]	0.62 [3.49]	0.60 [3.14]	0.63 [3.47]	0.85 [4.44]	0.39 [4.56]	0.41 [4.84]	0.43 [5.02]	0.36 [4.26]	0.35 [4.04]	0.31 [3.55]
Panel B: Portfolio average number of firms and market capitalization												
Size quintiles	CIA Quintiles					CIA Quintiles						
	Average n					Average market capitalization (\$10 ⁶)						
	(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)		
	(1)	364	366	366	365	361	32	30	29	29	31	
	(2)	101	101	101	102	101	56	56	55	56	55	
	(3)	72	73	72	72	72	100	99	97	98	100	
	(4)	62	62	62	62	61	219	227	217	224	217	
(5)	56	56	56	56	56	1334	1994	1758	2101	1435		

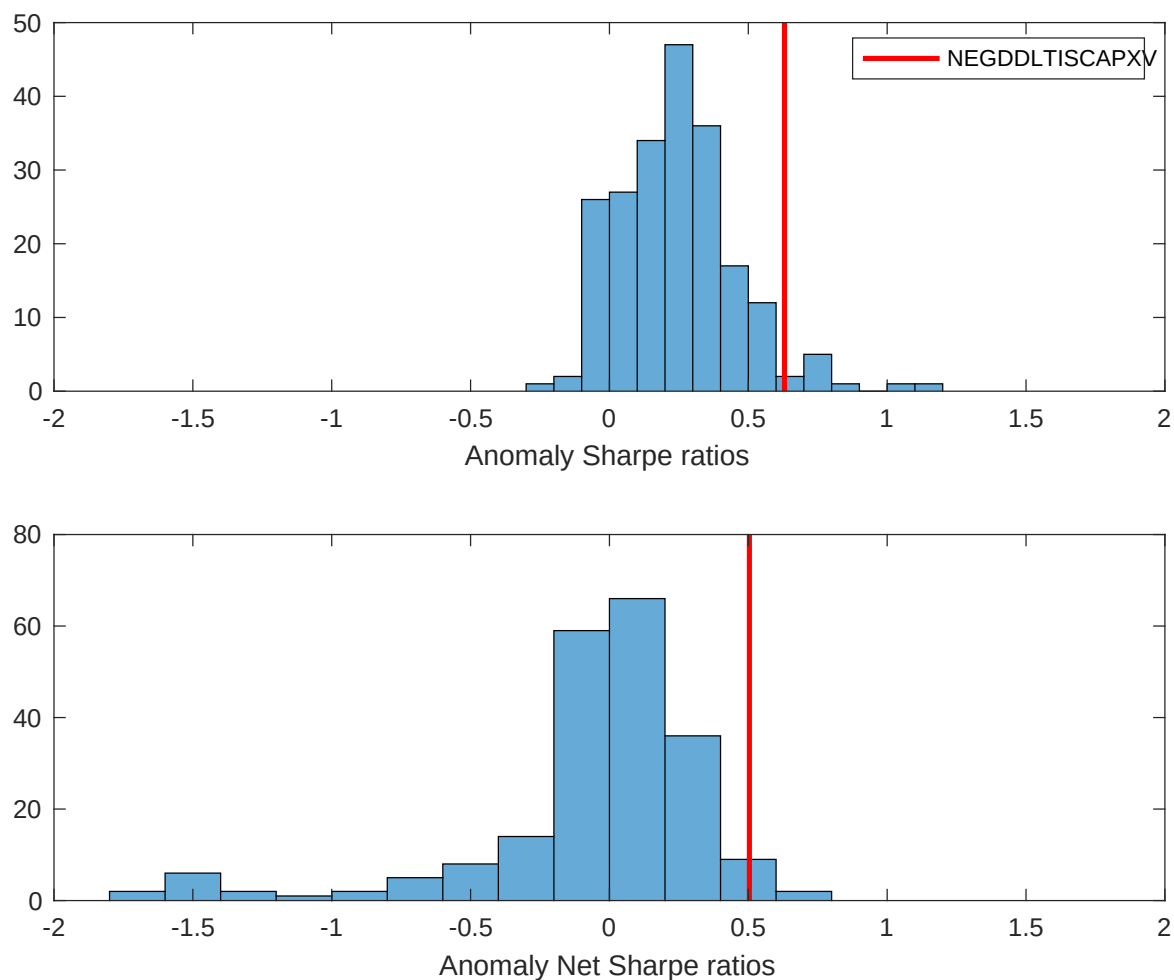


Figure 2: Distribution of Sharpe ratios.

This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the CIA with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

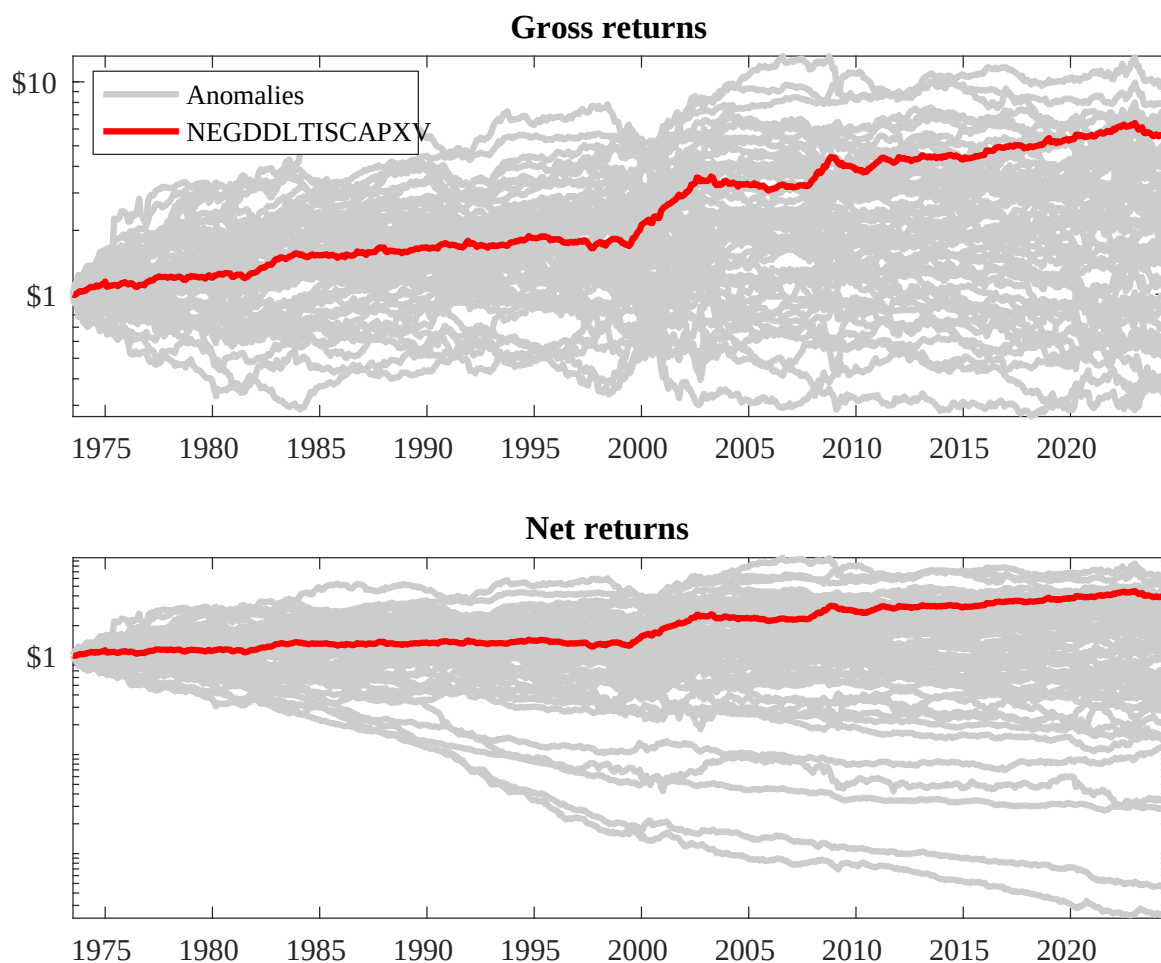
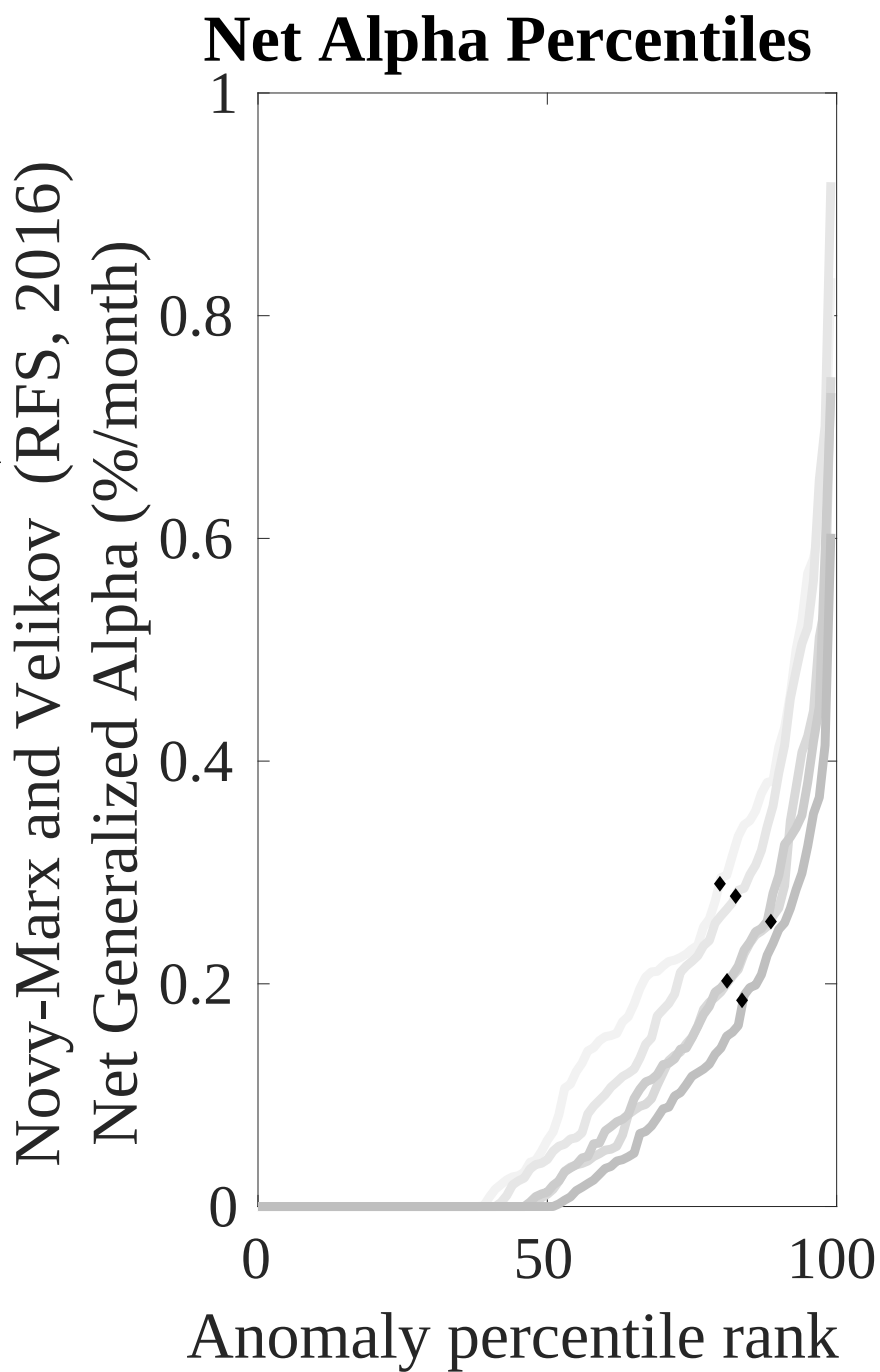
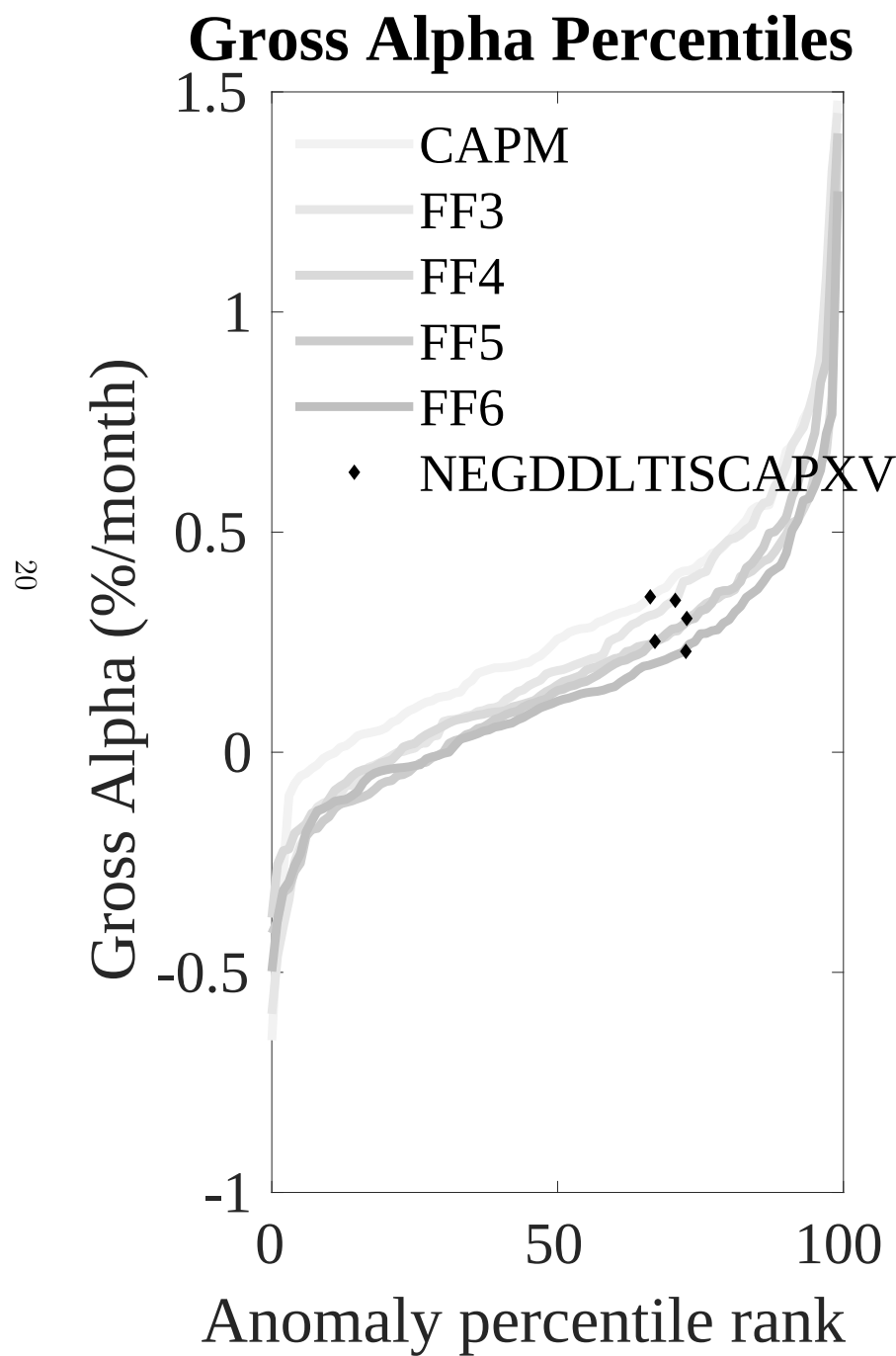


Figure 3: Dollar invested.

This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the CIA trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.



Novy-Marx and Velikov (RFS, 2016)

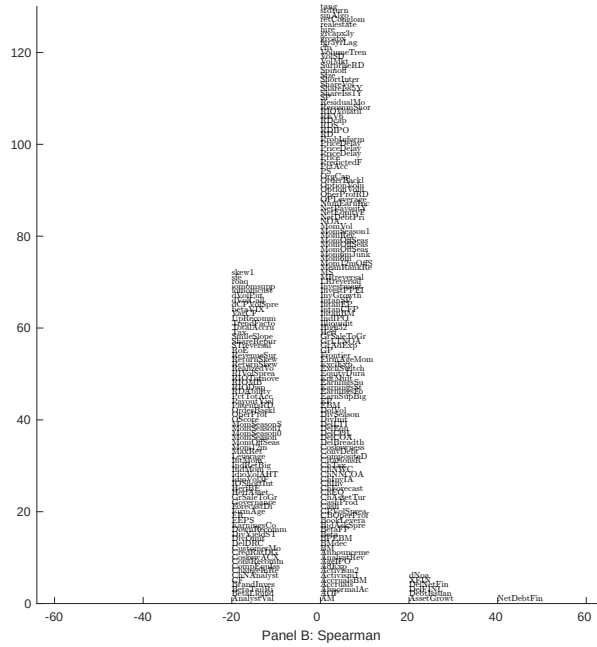
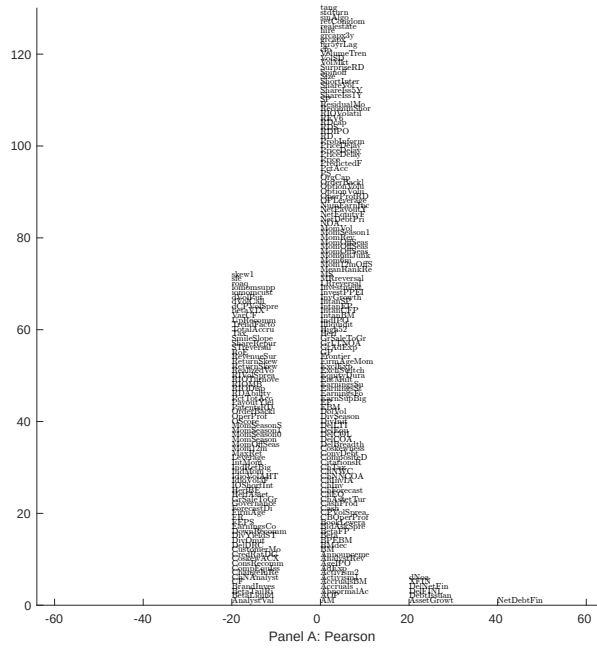


Figure 5: Distribution of correlations.

This figure plots a name histogram of correlations of 209 filtered anomaly signals with CIA. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

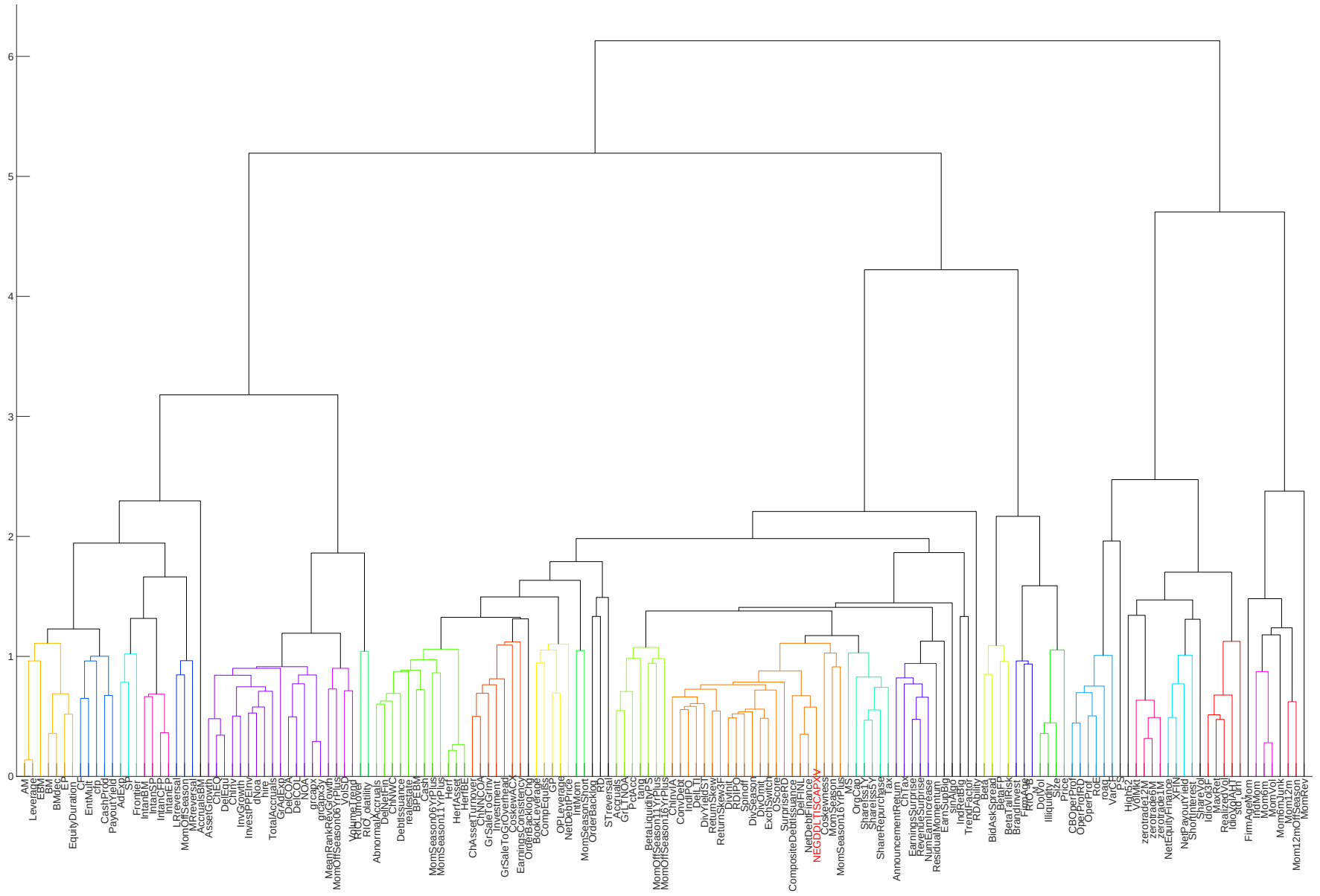


Figure 6: Agglomerative hierarchical cluster plot

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

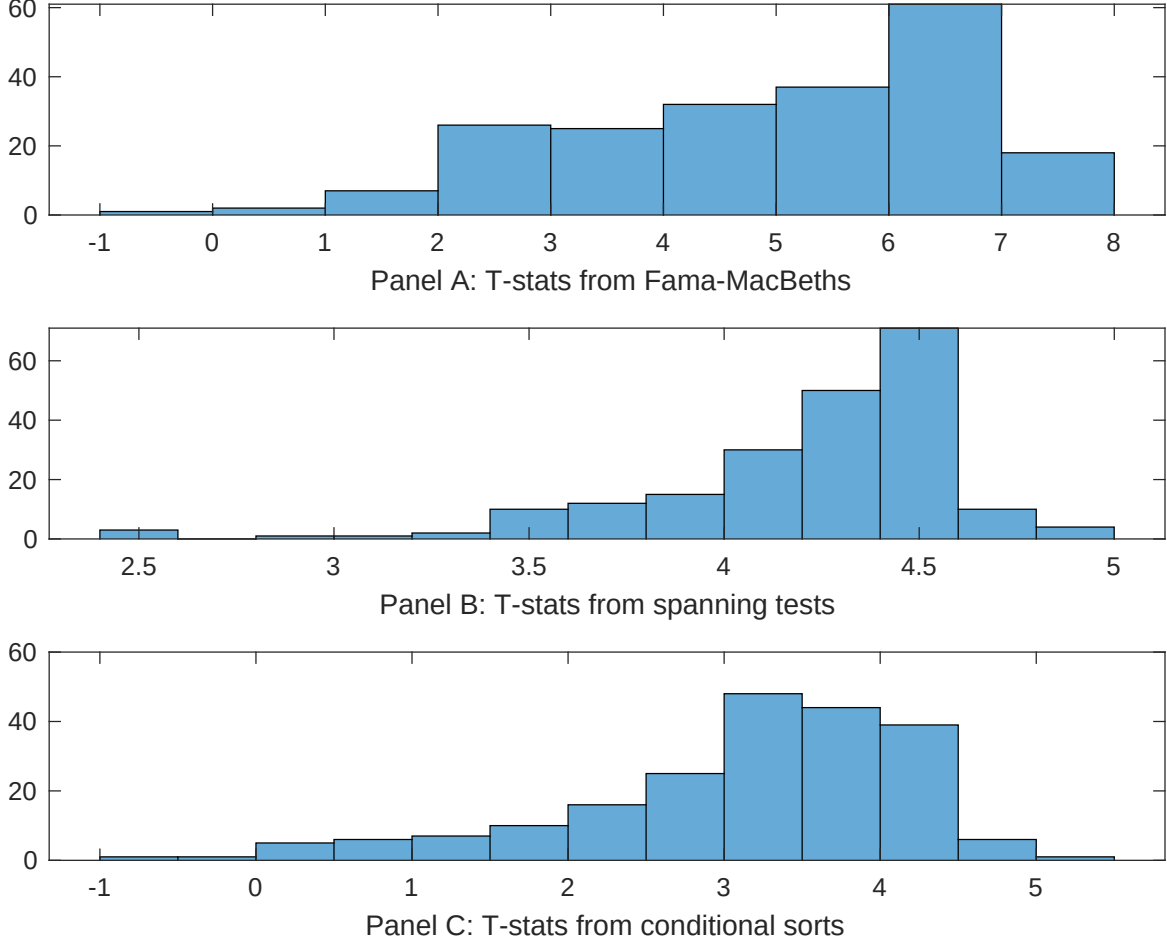


Figure 7: Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of CIA conditioning on each of the 209 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{CIA} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{CIA}CIA_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 209 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{CIA,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 209 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 209 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on CIA. Stocks are finally grouped into five CIA portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted CIA trading strategies conditioned on each of the 209 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on CIA. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{CIA}CIA_{i,t} + \sum_{k=1}^s \beta_{X_k}X_{i,t}^k + \epsilon_{i,t}$. The six most closely related anomalies, X , are Change in financial liabilities, Net external financing, Net debt financing, Asset growth, change in ppe and inv/assets, change in net operating assets. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 197306 to 202406.

Intercept	0.13 [5.36]	0.14 [5.69]	0.13 [5.33]	0.14 [5.78]	0.14 [5.74]	0.14 [5.66]	0.15 [6.02]
CIA	0.65 [2.53]	0.93 [3.38]	0.76 [2.70]	0.65 [2.33]	0.94 [3.54]	0.57 [2.10]	0.34 [1.20]
Anomaly 1	0.17 [8.90]						-0.63 [-1.53]
Anomaly 2		0.20 [6.45]					0.84 [1.60]
Anomaly 3			0.20 [8.97]				0.64 [1.13]
Anomaly 4				0.10 [8.51]			0.39 [2.01]
Anomaly 5					0.17 [7.53]		0.58 [2.43]
Anomaly 6						0.14 [9.77]	0.57 [3.39]
# months	612	612	612	612	612	612	612
$\bar{R}^2(\%)$	0	1	0	0	0	0	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the CIA trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{CIA} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies, X , are Change in financial liabilities, Net external financing, Net debt financing, Asset growth, change in ppe and inv/assets, change in net operating assets. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 197306 to 202406.

Intercept	0.21 [3.08]	0.21 [3.10]	0.21 [3.15]	0.22 [3.26]	0.22 [3.29]	0.22 [3.23]	0.21 [3.13]
Anomaly 1	13.41 [3.53]						10.93 [2.08]
Anomaly 2		10.68 [3.17]					6.85 [1.86]
Anomaly 3			13.60 [3.72]				5.32 [1.06]
Anomaly 4				7.16 [1.83]			5.35 [1.27]
Anomaly 5					5.60 [1.84]		4.38 [1.33]
Anomaly 6						1.36 [0.35]	-8.19 [-1.82]
mkt	-2.35 [-1.53]	-1.20 [-0.74]	-2.64 [-1.71]	-2.67 [-1.72]	-2.74 [-1.77]	-2.67 [-1.71]	-1.55 [-0.96]
smb	-0.65 [-0.27]	4.15 [1.62]	-0.48 [-0.20]	0.21 [0.09]	0.75 [0.32]	0.75 [0.32]	0.82 [0.30]
hml	-6.89 [-2.31]	-6.59 [-2.20]	-7.23 [-2.43]	-8.11 [-2.71]	-8.38 [-2.78]	-7.88 [-2.61]	-6.11 [-2.01]
rmw	5.31 [1.75]	0.14 [0.04]	5.02 [1.65]	6.54 [2.15]	6.62 [2.18]	6.54 [2.15]	0.82 [0.22]
cma	22.79 [4.86]	20.31 [4.05]	23.74 [5.16]	18.61 [2.84]	22.83 [4.44]	26.25 [4.89]	13.87 [2.05]
umd	2.57 [1.64]	3.66 [2.39]	2.86 [1.85]	3.88 [2.51]	3.72 [2.41]	3.70 [2.38]	2.77 [1.77]
# months	612	612	612	612	612	612	612
$\bar{R}^2(\%)$	12	12	13	11	11	11	14

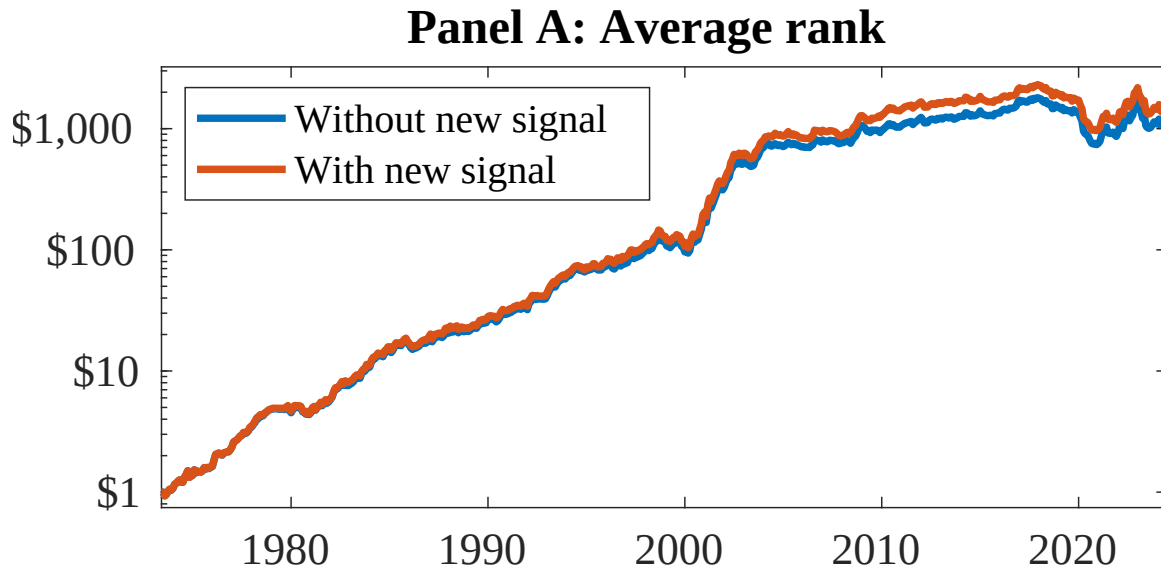


Figure 8: Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 155 anomalies. The red solid lines indicate combination trading strategies that utilize the 155 anomalies as well as CIA. Panel A shows results using "Average rank" as the combination method. See Section 7 for details on the combination methods.

References

- Bansal, R. and Yaron, A. (2004). Risks for the long run: A potential resolution of asset pricing puzzles. *Journal of Finance*, 59(4):1481–1509.
- Barro, R. J. (2006). Rare disasters and asset markets in the twentieth century. *Quarterly Journal of Economics*, 121(3):823–866.
- Campbell, J. Y. and Cochrane, J. H. (1999). By force of habit: A consumption-based explanation of aggregate stock market behavior. *Journal of Political Economy*, 107(2):205–251.
- Carhart, M. M. (1997). On persistence in mutual fund performance. *Journal of Finance*, 52:57–82.
- Chen, A. and Velikov, M. (2022). Zeroing in on the expected returns of anomalies. *Journal of Financial and Quantitative Analysis*, Forthcoming.
- Chen, A. Y. and Zimmermann, T. (2022). Open source cross-sectional asset pricing. *Critical Finance Review*, 27(2):207–264.
- Cochrane, J. H., Piazzesi, M., and Tuzel, S. (2023). Consumption-based asset pricing: Stock returns and consumption data. *Journal of Financial Economics*, 147(2):324–351.
- Cooper, M. J., Gulen, H., and Schill, M. J. (2008). Asset growth and the cross-section of stock returns. *Journal of Finance*, 63(4):1609–1651.
- Detzel, A., Novy-Marx, R., and Velikov, M. (2022). Model comparison with transaction costs. *Journal of Finance*, Forthcoming.
- Fama, E. F. and French, K. R. (1993). Common risk factors in the returns on stocks and bonds. *Journal of Financial Economics*, 33(1):3–56.

- Fama, E. F. and French, K. R. (2015). A five-factor asset pricing model. *Journal of Financial Economics*, 116(1):1–22.
- Fama, E. F. and French, K. R. (2018). Choosing factors. *Journal of Financial Economics*, 128(2):234–252.
- Gomes, J. F. and Schmid, L. (2021). Equilibrium asset pricing with leverage and default. *Journal of Finance*, 76(2):977–1018.
- Hou, K., Xue, C., and Zhang, L. (2015). Digesting anomalies: An investment approach. *Review of Financial Studies*, 28(3):650–705.
- Lyandres, E. and Michaely, R. (2016). The role of debt in the cross-section of equity returns. *Journal of Financial Economics*, 121(1):1–27.
- Merton, R. C. (1973). An intertemporal capital asset pricing model. *Econometrica*, 41(5):867–887.
- Novy-Marx, R. and Velikov, M. (2016). A taxonomy of anomalies and their trading costs. *Review of Financial Studies*, 29(1):104–147.
- Novy-Marx, R. and Velikov, M. (2023a). Assaying anomalies. *Working paper*.
- Novy-Marx, R. and Velikov, M. (2023b). Assaying anomalies. *Journal of Finance*, 78(3):1481–1521.
- Zhang, L. (2005). The value premium. *Journal of Finance*, 60(1):67–103.